

TSHI Sui Tiat(Xu Ruizhe)

Robotics, AI and Data Analysis | Mobile: (+65) 9353 0726 | Email: ruizhexu@u.nus.edu | [linkedin.com/in/ruizhexu422040](https://www.linkedin.com/in/ruizhexu422040)

PROFESSIONAL SUMMARY

MSc student in Smart Industries and Digital Transformation in tracks of Robotics and Automation with dual bachelor's training in foundation of Mathematical Science. Experience in statistical modelling, database development, computer vision research and computational simulation; seeking entry-level data analytics, AI, Robotics or digital transformation roles.

Strength: Enthusiastic, Fast Learner, Strong team player, Highly flexible, Resilient

CORE COMPETENCE

- Robotics and Computer Vision
- Data Analysis
- Agentic AI
- Attention to Detail
- Team Collaboration
- Project Leadership

EDUCATION

National University of Singapore (NUS)

Singapore

Master of Science in Smart Industries and Digital Transformation

August 2026 – January 2028(Expected)

- Graduate Certificate Track: Robotics and Automation (Expected)
- Relevant Coursework (in progress): Autonomous Systems, Machine Vision, Data Analysis in Sense-Making, Advanced Robotics, and Statistical Pattern Recognition.

Dalian University of Technology (DUT), Project 985 & 211

Dalian, China

Bachelor of Science in Foundations of Mathematical Science

September 2022 - June 2026

University of Leicester (UoL), UK

Dalian, China

Bachelor of Science in Mathematics

September 2022 - June 2026

- Second Class(Upper) Honours degree

University of Leicester (UoL), UK

Leicester, UK

Summer School

July 2024 - August 2024

- Improved English skills through General and Academic English courses, including Everyday English and poster presentation training; used advanced mathematical tools in lab sessions; learned about current advances in mathematics through lectures.

WORK EXPERIENCE

Yiku Business Management (Dalian) Co., Ltd.

Dalian, China

Data Analysis Intern

July 2024 - August 2024

- Database Maintenance:** Performed regular maintenance, including index optimization, removal of redundant data, and SQL query analysis to ensure system efficiency and stability.
- Key Outcomes:** Applied object-oriented principles, covering encapsulation, inheritance, and polymorphism, more rigorously in class design, preventing data corruption from improper interface use or errors; Improved communication and teamwork through collaboration, enhancing problem-solving and project execution.
- Object-Oriented Design:** Applied encapsulation, inheritance, and polymorphism in class design to improve code organisation and reduce the risk of errors caused by improper interface use.
- Team Collaboration:** Collaborated with team members on database development and maintenance, contributing to technical problem-solving and project delivery.

PROJECT EXPERIENCE

Vision II Football Penalty Data Analysis

Singapore

Group Project; Instructed by A.P. Vil Gopal

Aug. 2026 – Nov. 2026

- **Pipeline Development:** Built a reproducible Python workflow using OpenCV and pretrained YOLO11 models to process 39 penalty kicks from four 2018 FIFA World Cup shoot-outs and export structured feature datasets.
- **Pose-Based Feature Engineering:** Derived kicking-foot and run-up step estimates from hip-relative ankle trajectories and SciPy peak detection; incorporated frame-by-frame review to confirm step counts for all 39 kicks.
- **Motion Analysis:** Implemented camera-motion compensation for goalkeeper movement and dive features, and motion-constrained beam search to associate ball detections into image-plane trajectories.
- **Data Quality and Analysis:** Linked features to observed goal/no-goal labels; exported CSV datasets, diagnostic JSON and visual QA reports with confidence scores and review flags for uncertain cases.

Research on Out-of-Distribution Detection for Autonomous Driving Object Recognition

Dalian, China

Group Leader; Instructed by A.P. Jiankang Ren

Jan. 2024 - Jan. 2025

- **Description:** This project studies and applies advanced out-of-distribution detection methods, particularly Spatial-Temporal Unknown Distillation (STUD), to improve the robustness and safety of autonomous driving systems in complex scenarios.
- **STUD Method Application and Implementation:** Studied the STUD algorithm, which extracts unknown object features from natural video streams and regularises model decision boundaries to prevent overconfident classification of unknowns.
- **Extreme Weather Dataset Optimisation:** Enhanced existing public datasets (e.g., COCO) by augmenting data for extreme weather conditions (e.g., fog, rain, snow) where samples are limited.
- **Systematic Experimentation and Evaluation:** Conducted comparative experiments on COCO2017 and our optimized extreme weather dataset to evaluate performance, which improved out-of-sample detection **AUC-ROC by 15%** over the baseline and significantly reduced false detection rates, demonstrating its effectiveness.

Effect of vdW correction on the electronic properties of pentagonal PdSe₂

Dalian, China

Capstone Project; Instructed by A.P. Lizhao Liu

Jan. 2026 – May. 2026

- **Description:** Quantify vdW(van der Waals) correction impacts on pentagonal PdSe₂ nanoribbon electronic properties via DFT, and clarify the necessity of dispersion corrections for accurate bandgap prediction.
- **Model Construction:** Build 4 pristine nanoribbon topologies, plus central/lateral -F/-OH edge modified structures with full geometric relaxation.
- **Comparative Calculation:** Run parallel simulations with PBE, DFT-D3 and TS correction under identical computing parameters.
- **Artefact Diagnosis:** Compare band structures/DOS, identify spurious quasi-metallic errors from uncorrected PBE without dispersion forces.
- **Result Analysis:** Summarise bandgap convergence rule, reveal pentagonal lattice's defect tolerance and core modulation mechanism of vdW interactions.

Linear Statistics Model Project

Dalian, China

Individual Project; Instructed by Dr. Matias Ruiz

Mar. 2025 – May. 2025

- **Description:** This project develops a quantitative model relating welding parameters, covering current (A), voltage (V), electrode diameter (D), to weld joint fatigue life (R) via experimental design and statistical modelling, and identifies the optimal parameter combination for maximum fatigue resistance.
- **Experimental Design and Data Preparation:** Used response surface methodology (central composite or Box-Behnken design) to conduct experiments across varying levels of process parameters, and collected corresponding fatigue cycle data.
- **Statistical Model Construction and Significance Analysis:** Fitted a second-order polynomial model, and identified significant terms through hypothesis testing (e.g., t-tests, $p < 0.05$).
- **Model Diagnostics and Optimisation via Logarithmic Transformation:** Log-transformed ($\ln R$) R, and refitted the model to meet model assumptions such as homoscedasticity.
- **Model Interpretation and Process Parameter Optimisation:** Used the final $\ln R$ model to interpret parameter effects and determine the settings that maximise predicted fatigue life.

SKILLS AND LANGUAGES

- **Laboratory Techniques:** Semiconductor Properties Simulation & Analysis, Computer Vision Algorithm optimization
- **Skills:** Proficient in C, Java, Python, Matlab, R, Material Studio and Origin
- **Language:** Fluent in English (TOEFL IBT 93/120 and Writing 28/30) & Mandarin, Beginner of Japanese

ADDITIONAL INFORMATION

- **Leadership:** Foreign Affairs (responsible for venue management and planning); Vice Class Monitor in Bachelor Period
- **Personal Interests:** Piano, Level 10